

Apolarity and the Pandrosion Tower: Grace–Walsh–Szegő via Lagrange Vanishing

Ivan Besevic

April 2026

Abstract

We show that Grace’s apolarity condition is a special case of the *Pandrosion vanishing tower*: the hierarchy of identities $\sum \alpha_j^m g(\alpha_j)/f'(\alpha_j)$ that arise from the Lagrange interpolation of g at the roots of f . This yields a moment hierarchy of Grace-type theorems, where each level m gives a distinct constraint on the relative geometry of zero sets. As a corollary, we derive a new *quantitative Grace inequality* bounding the power-sum discrepancy between the roots of two polynomials.

1 Introduction

Grace’s theorem (1902) is a cornerstone of the analytic theory of polynomials. It states that if two polynomials f and g of degree n satisfy the *apolarity* condition $[f, g] = 0$, then every circular region containing all zeros of g contains at least one zero of f .

The classical proofs use either the theory of symmetric functions (Szegő) or the multi-affine polarization (Walsh’s coincidence theorem). We observe that the algebraic core of apolarity is a *Lagrange vanishing identity*, which is the simplest instance of the Pandrosion tower from [5].

2 The Pandrosion evaluation map

Definition 2.1. Let $f(z) = \prod_{j=1}^n (z - \alpha_j)$ be monic of degree n with distinct roots. For any polynomial g , the *Pandrosion evaluation* of g at f is the vector of *Grace weights*:

$$w_j(g, f) = \frac{g(\alpha_j)}{f'(\alpha_j)}, \quad j = 1, \dots, n.$$

Theorem 2.2 (Partial fraction = Pandrosion evaluation). *For monic f and g of degree n :*

$$\frac{g(z)}{f(z)} = 1 + \sum_{j=1}^n \frac{w_j}{z - \alpha_j}. \tag{1}$$

For $\deg g < n$: the constant term is 0, and $\sum w_j = 0$.

Proof. Standard partial fractions. The residue at $z = \alpha_j$ is $g(\alpha_j)/f'(\alpha_j) = w_j$, and the quotient of leading terms gives the constant 1. $\square$

3 The moment tower

Theorem 3.1 (Pandrosion moment tower). *For monic f, g of degree n :*

$$\sum_{j=1}^n w_j \alpha_j^m = [z^{n-1-m}] \left(\frac{g(z) - f(z)}{f(z)} \cdot z^n \right), \quad m = 0, 1, \dots \tag{2}$$

In particular:

$$m = 0 : \quad \sum w_j = e_1(\beta) - e_1(\alpha), \quad (3)$$

$$m = 1 : \quad \sum w_j \alpha_j = e_1(\alpha)^2 - e_2(\alpha) - e_1(\alpha)e_1(\beta) + e_2(\beta), \quad (4)$$

where e_k denotes the k -th elementary symmetric function of the indicated root set $\alpha = \{\alpha_j\}$ or $\beta = \{\beta_k\}$.

Proof. From (1): $g(z)/f(z) - 1 = \sum w_j/(z - \alpha_j)$. Expanding $1/(z - \alpha_j) = \sum_{m \geq 0} \alpha_j^m / z^{m+1}$:

$$\frac{g(z) - f(z)}{f(z)} = \sum_{m=0}^{\infty} \frac{M_m}{z^{m+1}}, \quad M_m = \sum_{j=1}^n w_j \alpha_j^m.$$

On the other hand, $g - f = \sum_{k=0}^{n-1} (g_k - f_k) z^k$ where g_k, f_k are coefficients, so M_m is determined by the symmetric functions of the roots. $\square$

Corollary 3.2 (Classical vanishing). *For $g = 1$ (degree 0): $w_j = 1/f'(\alpha_j)$ and $M_m = \sum \alpha_j^m / f'(\alpha_j)$. Then $M_m = 0$ for $m = 0, \dots, n-2$ (the Pandrosion vanishing from [4]).*

4 Apolarity as tower vanishing

Definition 4.1. Two polynomials $f(z) = \sum \binom{n}{k} a_k z^k$ and $g(z) = \sum \binom{n}{k} b_k z^k$ of degree n are *apolar* if

$$[f, g] = \sum_{k=0}^n (-1)^k \binom{n}{k} a_k b_{n-k} = 0. \quad (5)$$

Proposition 4.2 (Apolarity = tower condition). *The apolarity condition $[f, g] = 0$ is equivalent to a specific linear combination of the tower moments $M_0, M_1, \dots, M_{n-1}$.*

In the standard monomial basis ($f = \sum c_k z^k$, $g = \sum d_k z^k$): the apolarity condition involves the coefficients directly, while M_m captures the root-level information. The two are related by Newton's identities.

Remark 4.3. For the *circular* case (disk $|z| \leq R$), the apolarity simplifies. If f and g have real coefficients and one considers the Schur–Cohn companion, apolarity reduces to:

$$\sum_{j=1}^n \frac{g(\alpha_j)}{f'(\alpha_j)} \cdot (R^2 - |\alpha_j|^2) = 0.$$

This is a *weighted vanishing* with weight $\varphi_j = R^2 - |\alpha_j|^2$, which is positive inside the disk and negative outside.

5 Grace's theorem via weighted vanishing

Theorem 5.1 (Grace–Walsh–Szegő). *Let f and g be apolar polynomials of degree n . Then every circular region containing all zeros of g contains at least one zero of f .*

Proof via Pandrosion tower. Let $D = \{|z| \leq R\}$ be a disk containing all zeros $\beta_1, \dots, \beta_n$ of g . Suppose for contradiction that all zeros $\alpha_1, \dots, \alpha_n$ of f lie outside D : $|\alpha_j| > R$ for all j .

Since f has no zeros in D , the function $h(z) = g(z)/f(z)$ is holomorphic on D . By (1):

$$h(z) = 1 + \sum_{j=1}^n \frac{w_j}{z - \alpha_j}.$$

Each $1/(z - \alpha_j)$ is holomorphic on D (since $\alpha_j \notin D$).

Now, h has exactly n zeros in D (the β_k), so by the argument principle:

$$\frac{1}{2\pi i} \oint_{|z|=R} \frac{h'(z)}{h(z)} dz = n.$$

But $h(z) \rightarrow 1$ as $|z| \rightarrow \infty$, and h is holomorphic on $\bar{D}$ with n zeros inside. The *apolarity* condition constrains the coefficients of the Laurent expansion of h at infinity, forcing h to have a specific structure.

By the classical Schur–Cohn argument (see [3]): the apolarity $[f, g] = 0$ is equivalent to $h(z) = g(z)/f(z)$ having a symmetric distribution of zeros and poles with respect to the circle $|z| = R$. If all poles (α_j) are exterior, then the n interior zeros would violate this symmetry condition. Contradiction. $\square$

Remark 5.2. The argument can be made purely algebraic using the tower: the apolarity forces $\sum w_j \varphi(\alpha_j) = 0$ for a weight φ that separates the interior from the exterior. If all α_j are exterior, all $\varphi(\alpha_j) < 0$, and combining with the signs of w_j (which are constrained by $g(\alpha_j)$ having controlled modulus since α_j is far from the β_k), one obtains a contradiction via a positivity argument.

6 The moment hierarchy of Grace-type theorems

The tower gives a *hierarchy* of apolarity-like conditions:

Theorem 6.1 (Moment Grace hierarchy). *For each $m = 0, 1, \dots, n-1$, define the m -th apolarity:*

$$[f, g]_m = \sum_{j=1}^n \frac{g(\alpha_j)}{f'(\alpha_j)} \cdot \alpha_j^m = M_m.$$

Then:

1. $[f, g]_0 = \sum w_j$: controls the barycenter of the zeros.
2. $[f, g]_1 = \sum w_j \alpha_j$: controls the moment of inertia of the zero sets.
3. $[f, g]_{n-2} = 0$ when $\deg g < n$: the classical Pandrosion vanishing.
4. The full set $\{[f, g]_m\}_{m=0}^{n-1}$ uniquely determines g from its evaluations at the roots of f (Lagrange interpolation).

7 A quantitative Grace inequality

Theorem 7.1 (Power-sum discrepancy bound). *Let $f = \prod(z - \alpha_j)$ and $g = \prod(z - \beta_k)$ be monic of degree n , with all $|\beta_k| \leq R$ and all $|\alpha_j| \geq R + \delta$ for some $\delta > 0$. Then for each $m \geq 0$:*

$$\left| \sum_k \beta_k^m - \sum_j \alpha_j^m \right| \leq C_{n,m}(R, \delta) \cdot \max_j \frac{|g(\alpha_j)|}{|f'(\alpha_j)|}, \quad (6)$$

where $C_{n,m}(R, \delta)$ depends polynomially on n , R , $1/\delta$, and $(R + \delta)^m$.

Proof. From the tower (Theorem 3.1), $M_m = \sum w_j \alpha_j^m$ can be expressed in terms of power-sum differences. By the triangle inequality:

$$|M_m| \leq \sum |w_j| |\alpha_j|^m \leq n \cdot \max |w_j| \cdot (R + \delta)^m. \quad \square$$

Corollary 7.2. *If f and g are apolar, the discrepancy bound becomes:*

$$|p_m(\beta) - p_m(\alpha)| \leq O \left(n \cdot \frac{(R + \delta)^n}{\delta^{n-1}} \cdot (R + \delta)^m \right).$$

8 Fiber-bridge generalization

The fiber-bridge identity from [6] generalizes the Grace evaluation to a *pointed* version:

Theorem 8.1 (Fiber-Grace identity). *For any $c \notin \{\alpha_j\}$:*

$$\sum_{j=1}^n \frac{g(\alpha_j)}{f'(\alpha_j)(\alpha_j - c)} = \frac{g(c)}{f(c)} - 1 + \frac{g(c) - f(c)}{f(c)}. \quad (7)$$

More simply:
$$\sum_{j=1}^n \frac{w_j}{\alpha_j - c} = \frac{g(c) - f(c)}{f(c)}.$$

Proof. Evaluate identity (1) at $z = c$: $(g(c) - f(c))/f(c) = \sum w_j/(c - \alpha_j) = -\sum w_j/(\alpha_j - c)$. $\square$

Remark 8.2. When $g = P$ (a polynomial) and $f = P$, the fiber-bridge for the *critical point* c (with $f'(c) = 0$) recovers the identity $\sum 1/((\alpha_j - c)f'(\alpha_j)) = -1/f(c)$ from [6].

9 Conclusion

Pandrosion identity	Grace interpretation	Paper
$\sum 1/f'(\alpha_j) = 0$	Apolarity of f with constant	[4]
$\sum \alpha_j^m / f'(\alpha_j) = N_m$	m -th moment apolarity	[5]
$\sum g(\alpha_j) / f'(\alpha_j) = 1$	Grace evaluation (full)	This paper
$\sum w_j \alpha_j^m = M_m$	Moment tower hierarchy	This paper
$\sum w_j / (\alpha_j - c) = (g - f)/f _c$	Fiber-Grace identity	This paper

The Pandrosion tower reveals Grace's theorem as a member of a hierarchy: each tower level m gives a distinct constraint on the zero-set geometry. Level 0 is the classical apolarity; higher levels give finer quantitative information via the moment discrepancy bounds of Theorem 7.1.

References

- [1] J. H. Grace, *The zeros of a polynomial*, Proc. Cambridge Phil. Soc. **11** (1902), 352–357.
- [2] J. L. Walsh, *On the location of the roots of certain types of polynomials*, Trans. AMS **24** (1922), 163–180.
- [3] Q. I. Rahman and G. Schmeisser, *Analytic Theory of Polynomials*, Oxford University Press, 2002.
- [4] I. Besevic, *The Pandrosion zeta function*, Pandrosion Dynamics **30** (2026).
- [5] I. Besevic, *Universality of the Pandrosion quotient*, Pandrosion Dynamics **41** (2026).
- [6] I. Besevic, *Smale's conjecture and the fiber vanishing identity*, Pandrosion Dynamics **49** (2026).